

07c — Functional Testing

Key Concepts

Term	Definition
Functional Test	Verifies a feature works according to business requirements from user's perspective
Black Box	Tests only inputs and outputs — doesn't care about internal code
E2E Test	End-to-end test — often synonymous with functional testing
User Story	Business requirement that defines what functional tests should verify
Browser Automation	Tools (Selenium, Cypress, Playwright) that simulate real user interactions

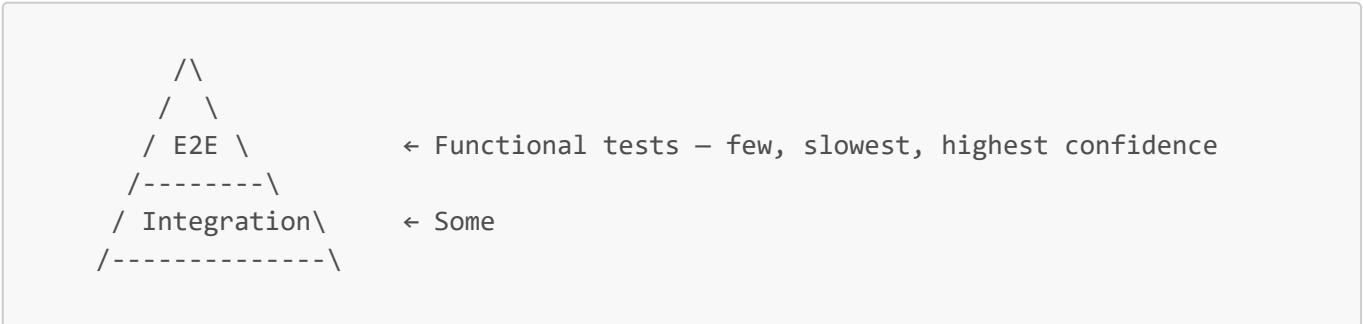
How It Works

- 1. Define expected behavior from business requirements / user stories
- 2. Automate a real user flow through the UI or API
- 3. Assert the user-facing outcome (not internal state)
- 4. Run before releases — not on every commit

The Three Testing Levels Compared

	Unit	Integration	Functional
Perspective	Developer	Developer	User / Business
Tests	Code logic	Component connections	User-facing behavior
Dependencies	Mocked	Real	Real (full system)
Speed	Fastest	Moderate	Slowest
Written by	Developers	Developers	QA / Business Analysts
Pyramid position	Bottom (many)	Middle (some)	Top (few)

Testing Pyramid



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 /   Unit Tests   \   ← Many, fastest
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```

Industry Examples

- **Uber** — functional test for ride booking: user logs in → enters destination → confirms → verifies driver assigned, ETA shown, fare displayed
- **Amazon** — functional test for checkout: add to cart → enter address → pay → verify order confirmation email received
- **Netflix** — functional test for playback: log in → select title → press play → verify video starts within 3 seconds

Tools

Tool	Notes
Selenium	Oldest, most widely supported browser automation
Cypress	Modern, developer-friendly, fast feedback
Playwright	Supports Chromium, Firefox, Safari — built by Microsoft

Interview Questions

- Q1: What is functional testing?** Testing whether a feature meets business requirements from the user's perspective. Treats the system as a black box — only the outcome matters, not how the code works internally.
- Q2: How is functional testing different from unit and integration testing?** Unit tests verify code logic in isolation. Integration tests verify components connect correctly. Functional tests verify the user gets the expected outcome — defined by business requirements, not code structure.
- Q3: Who writes functional tests?** Typically QA engineers or business analysts. Test cases come from user stories and business requirements, not from reading the code.
- Q4: What tools are used for functional testing?** Selenium, Cypress, and Playwright — they automate a real browser, simulating user clicks and interactions end-to-end.
- Q5: Where does functional testing sit in the Testing Pyramid?** At the top — fewest in number, slowest to run, but highest confidence that the product works from a user's perspective.